No.

Date:

Sistem Persamaan Linear – Matriks

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1A TTE

1 a Metode Cramer

$2x + 3y + z = -11$

$2x - 3y + z = -17$

$2x + 3y - z = -3$

Penyelesaian :

Determinan D, Dx, Dy, dan Dz diperoleh :

- $D = \begin{bmatrix} 2 & 3 & 1 & 2 & 3 \\ 2 & -3 & 1 & 2 & -3 \\ 2 & 3 & -1 & 2 & 3 \end{bmatrix}$

$= (2 \times (-3) \times (-1)) + (3 \times 1 \times 2) + (1 \times 2 \times 3) - (3 \times 2 \times (-1)) - (2 \times 1 \times 3) - (1 \times (-3) \times 2)$

$= (6 + 6 + 6 + 6 - 6 + 6)$

$D = 24$

- $Dx = \begin{bmatrix} -11 & 3 & 1 & -11 & 3 \\ -17 & -3 & 1 & -17 & -3 \\ -3 & 3 & -1 & -3 & 3 \end{bmatrix}$

$= ((-11) \times (-3) \times (-1)) + (3 \times 1 \times -3) + (1 \times (-17) \times 3) - (3 \times (-17) \times (-1) - ((-11) \times 1 \times 3)$

$- (1 \times (-3) \times (-3)$

$= (-33 - 9 - 51 - 51 + 33 - 9)$

$Dx = -120$

- $Dy = \begin{bmatrix} 2 & -11 & 1 & 2 & -11 \\ 2 & -17 & 1 & 2 & -17 \\ 2 & -3 & -1 & 2 & -3 \end{bmatrix}$

$= (2 \times (-17) \times (-1) + ((-11) \times 1 \times 2) + (1 \times 2 \times (-3)) - ((-11) \times 2 \times (-1)) - (2 \times 3 \times (-3) - (1 \times (-17) \times 2$

$= 34 - 22 - 6 - 22 + 6 + 34$

$= 24$

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$$D_z = \begin{bmatrix} 2 & 3 & -11 & 2 & 3 \\ 2 & -3 & -17 & 2 & -3 \\ 2 & 3 & -3 & 2 & 3 \end{bmatrix}$$

$= (2\times(-3)\times(-3)) + (3\times(-17)\times 2) + (-11\times 2\times 3) - (3\times 2\times(-3)) - (2\times(-17)\times 3) - ((-11)\times(-3)\times 2)$

$= 18 - 102 - 66 + 18 + 102 - 66$

$= -96.$

$\Rightarrow x = \frac{D_x}{D} = \frac{-120}{24} = -5$

$\Rightarrow y = \frac{D_y}{D} = \frac{24}{24} = 1$

$\Rightarrow z = \frac{D_z}{D} = \frac{-96}{24} = -4$

maka $x = -5$; $y = 1$; $z = -4$

2 a Metode Invers Matriks

$4x - y + 2z = 15$

$x + 2y + 3z = 9$

$2x + y - z = 0$

Penyelesaian :

$$\begin{bmatrix} 4 & -1 & 2 \\ 1 & 2 & 3 \\ 2 & 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 15 \\ 9 \\ 0 \end{bmatrix}$$

- Mencari Determinan

$$\begin{bmatrix} 4 & -1 & 2 & 4 & -1 \\ 1 & 2 & 3 & 1 & 2 \\ 2 & 1 & -1 & 2 & 1 \end{bmatrix}$$

$= (4\times 2\times(-1) + (-1)\times 3\times 2) + (2\times 1\times 1) - ((-1)\times 1\times(-1)) - (4\times 3\times 1) - (2\times 2\times 2)$

$= -8 - 6 + 2 - 1 - 12 - 8$

$= -33$

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⇒ Mencari Adjoin :

$$\begin{bmatrix} -2-3 & 1-2 & -3-4 \\ -1-6 & -4-4 & 12-2 \\ 1-4 & 4-(-2) & 8-(-1) \end{bmatrix} = \begin{bmatrix} -5 & -1 & -7 \\ -7 & -8 & 10 \\ -3 & 6 & 9 \end{bmatrix}$$

$$Adj = \begin{bmatrix} -5 & 1 & -7 \\ 7 & -8 & -10 \\ -3 & -6 & 9 \end{bmatrix}$$

$$A^{-1} = \frac{1}{det} \cdot Adj$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \frac{1}{-33} \begin{bmatrix} -5 & 1 & -7 \\ 7 & -8 & -10 \\ -3 & -6 & 9 \end{bmatrix} \begin{bmatrix} 15 \\ 9 \\ 0 \end{bmatrix}$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \frac{1}{-33} \begin{bmatrix} -66 \\ 33 \\ -99 \end{bmatrix}$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$$ } maka $x = 2$; $y = -1$; $z = 3$

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